Problem 1. Prove that n^2 + n is even for every integer n.

Case 1: n is even, so n = 2k for some integer k. Then n^2 + n = 4k^2 + 2k = 2(2k^2 + k), which is even.

Case 2: n is odd, so n = 2k + 1 for some integer k. Then n^2 + n = 4k^2 + 4k + 1 + 2k + 1 = 2(2k^2 + 3k + 1), which is even.

Every integer is even or odd, so n^2 + n is even for every integer n.